EPSR-test-ex73 sdf water-water (identical groups), chained 6 times, with mapgr

Program(s): sdf, mapgr | chained passes: 6 | verification: ALL IDENTICAL (all 6 passes)

What this case tests

Exercises `sdf` [1], the variant of `sharm` in which molecules are defined by atom types and distance criteria, so that fragments or groups spanning molecules can be analysed; here whole water molecules, accumulated over a chained run and mapped with `mapgr`.

Code exercised

Fortran (original, fortran_src/):

- `sdf_rev2.f`: `run_sdf` (keywords), `sdf` (atom-group search by type and distance, pair loop, averaged groups)
- `sharm_routines_rev1.f`: `get_l_and_n_values`, `form_cgcoeffs`, `form_eulerangles`, `form_dlmn`, `form_sum_prod_dlmn_cgcoeff`
- `auxiliary_routines.f`: `dlm`, `cgcoef`, `rotateeuler`; `read_and_save_files.f`: `outgofzs_sharm` (`.h0n` files, averaging); `process_symbols.f`: `check_symbol`
- `mapgrquad.f`: `run_mapgr`; `plotsharm_routines.f`: `smoosharm`, `sumoverm2`; `plot_routines.f`: `find_plot_par`

Rust (port, rust/epsr-core/src/):

- `tools/sdf.rs`: `run`, `sdf`, `check_symbol_extra`
- `tools/sharm_routines.rs`: `Sharm` methods with the Fortran names (incl. `outgofzs_sharm`), `dlm`, `cgcoef`, `form_eulerangles`, `rotateeuler`
- `tools/mapgr.rs`: `run`; `tools/sharm_routines.rs`: `Coeffs::read`, `Coeffs::angular`, `smoosharm`, `sumoverm2`, `check_symbol_mapgr`; `output.rs`: `w_int`

Physical system

The EPSR-test configuration `200DMSO400Water.ato`: 200 DMSO (atoms S, C, C, O, 6 H) and 400 water molecules (O1, H1, H1; SPC/E parameters [2]), 3200 atoms at 300 K in a cubic box of side 33.12 Å, a model of the DMSO-water mixtures of Soper and Luzar [3]. `sdf` computes the same spherical-harmonic coefficients [4,5] as `sharm`, but its "molecules" are groups of atoms found by atom types and pair distances (first atom = centre). Groups O1 H1 H1 with pair distances 0.976, 0.976, 1.54 Å ±20 %, axes as for water in `sharm` (z to the H-H midpoint, y from H 2); identical central and second groups (the second-group distances and axes in the file are overwritten by `check_symbol`); l1, l2 = 0-2, l = 0-4, n steps 2 (38 coefficients). `mapgr` rebuilds g(r) along chosen directions and relative orientations from the coefficients (grtype 1 radial, 2 spatial, 3 full orientational).

Inputs

Input files written by hand from the keyword tables (`setup_symbols_*`), one `keyword value` per line, ended by `q`; `200DMSO400Water.ato` copied from EPSR-test. `water.SDF.dat` (nr 100, rmax 10, `averagemol y`); `water.mapgr.txt`: two full orientational views.

How the accuracy was verified

`commands.txt` (2 program calls, repeated 6 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. A run-time FORMAT in the original source (`sharm_rev2.f`/`sdf_rev2.f`, and `sharm_diffuse` for `epsr`) lacks a comma after the Hollerith `1h(` and is rejected by current gfortran, so the reference build inserts that comma (originals kept as `*.orig`); the port behaves like the patched program. Result of the final regression run of the complete port: every file identical in all 6 passes.
Six passes: calculation in pass 1 and in pass 6 (`nsumt 5`, averaged with the stored `.h01`), final `nsumt 6`. Compared: `.SDF.h01`, both xyz files, updated `.SDF.dat` and `.mapgr.g01` after every pass. Reference build: `sharm`/`sdf` were rebuilt with one comma inserted in the run-time title format (`'(1h(,'//il1fmt`), since gfortran 15.2 otherwise stops with "Missing comma between descriptors" before writing the coefficients; all other programs are the unmodified reference binaries. Also masked: characters 31-45 of the `*.mapgr.g01` data lines, the error column of `g_centres(r)`, which `mapgrquad.f` never sets (uninitialised in the Fortran, 0 in the port).

Notes

Fortran quirk reproduced: the remaining (H-H) distance is checked with `zdiff` = `ytemp(i2) - ztemp(i1)`, so according to the README only 235 of the 400 waters are accepted as groups.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] A. K. Soper, "Orientational correlation function for molecular liquids: The case of liquid water", J. Chem. Phys. 101, 6888-6901 (1994).
[5] C. G. Gray and K. E. Gubbins, Theory of Molecular Fluids, Vol. 1: Fundamentals, Clarendon Press, Oxford (1984).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.